Gary Cornwall

Senior Research Economist, U.S. Bureau of Economic Analysis

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Research Interests

Econometrics; causal inference; time series and forecasting; machine learning and AI evaluation; Bayesian, spatial, and network methods; urban and real estate economics.

Appointments and Affiliations

- **Senior Research Economist**, U.S. Bureau of Economic Analysis, August 2017-present.
- **Adjunct Professor of Economics**, George Washington University, January 2020-present.
- **Member**, H.O. Stekler Research Program on Forecasting, 2024-present.

Education

University of Cincinnati, Carl H. Lindner College of Business

- Ph.D., Economics, 2017.
- M.A., Economics, 2012.
- B.A., Economics, 2010.

Journal Articles

1. Chen, Jeff, Gary Cornwall, and Scott Wentland. "It's the Smell: How Resolving Uncertainty about Local Disamenities Can Affect the Housing Market." *Land Economics* 102(1) (2026): 49-75. DOI
2. Bianco, Timothy, Gary Cornwall, and Beau Sauley. "Financial Reform and Mortgage Lending by Systemically Important Financial Institutions." *Economics Letters* 256 (2025): 112633. DOI
3. Cornwall, Gary, and Marina Gindelsky. "Nowcasting Distributional National Accounts for the United States: A Machine Learning Approach." *AEA Papers and Proceedings* 115 (2025): 79-84. DOI
4. Cornwall, Gary, and Beau Sauley. "Indirect Effects and Causal Inference: Reconsidering Regression Discontinuity." *Journal of Spatial Econometrics* 2 (2021), article 8.
5. Strawn, J. R., J. A. Mills, G. J. Cornwall, S. A. Mossman, S. T. Varney, B. R. Keeshin, and P. E. Croarkin. "Buspirone in Children and Adolescents with Anxiety: A Review and Bayesian Analysis of Abandoned Randomized Controlled Trials." *Journal of Child and Adolescent Psychopharmacology* 28(1) (2018): 2-9.
6. Cornwall, Gary, and Olivier Parent. "Embracing Heterogeneity: The Spatial Autoregressive Mixture Model." *Regional Science and Urban Economics* 64 (2017): 148-161.
7. Strawn, J. R., E. T. Dobson, J. A. Mills, G. J. Cornwall, D. Sakolsky, B. Birmaher, S. N. Compton, J. Piacentini, J. T. McCracken, G. S. Ginsburg, P. C. Kendall, J. T. Walkup, A. M. Albano, and M. A. Rynn. "Placebo Response in Pediatric Anxiety Disorders: Results from the Child/Adolescent Anxiety Multimodal Study (CAMS)." *Journal of Child and Adolescent Psychopharmacology* 27(6) (2017): 501-508.

Professional and Agency Service

- Board Member, Society of Government Economists, 2025-present.
- Member, AI Governance Board, U.S. Bureau of Economic Analysis, 2025-present; representative of the Office of the Chief Economist.
- Conference Coordinating Committee, Society of Government Economists Conference, 2026.
- BEA Organizing Committee, FCSM Conference, 2026.

Book

Chen, Jeffrey C., Edward A. Rubin, and Gary J. Cornwall. *Data Science for Public Policy.* Springer International Publishing, 2021.

Book Chapters

1. Cornwall, Gary J., Jeffrey A. Mills, Beau A. Sauley, and Huibin Weng. "Predictive Testing for Granger Causality via Posterior Simulation and Cross-validation." In *Topics in Identification, Limited Dependent Variables, Partial Observability, Experimentation, and Flexible Modeling: Part A*, *Advances in Econometrics* 40A (2019): 275-292.
2. Cornwall, Gary, Changjoo Kim, and Olivier Parent. "At the Frontier Between Local and Global Interactions in Regional Sciences." In *Regional Research Frontiers, Volume II*, edited by Randall Jackson and Peter Schaeffer, 141-149. Springer, 2017.

Current Projects and Working Papers

1. **It's About Time (Series): A Simple Correction for Difference-in-Differences.** With Nicholas Brown and Scott Wentland.
2. **The Seasons They Are A-Changin': A Century of Definitions and a Way Forward.** With Carter Bryson. BEA Working Paper WP2026-14, 2026.
3. **Whose Hand Do You Prefer? The Steerability of Machine Aesthetic Judgment.** Maya Ferrín, Jeffrey C. Chen, and Gary Cornwall. Working paper, 2026. Accepted for presentation, NeurIPS Creative AI Track 2026.
4. **Alloy Inference: Tests of a Single Null.**
5. **For What It's Worth: Measuring Land Value in the Era of Big Data and Machine Learning.** With Scott Wentland and Jeremy Moulton.
6. **Roots from Trees: A Machine Learning Approach to Unit Root Detection.** With Jeffrey C. Chen and Beau Sauley.
7. **The Dark Side of the Moon: Searching for the Other Half of Seasonality.** With Jeffrey C. Chen.
8. **When the Levy Breaks: How Failing Renewal School Levies Impacts Housing Prices.** With Jeremy Moulton, Beau Sauley, and Scott Wentland.
9. **Monetary Policy and the Housing Market: Evidence from National Microdata.** With Jeremy Moulton and Scott Wentland.
10. **Segment Reporting and Financial Statement Comparability: An Examination of Privately Reported Industry Data and Publicly Reported Segment Data.** With Matthew Cobabe, Bret Johnson, Kelly Wentland, and Scott Wentland.
11. **Aggregation Bias and Input-Output Regionalization: Detail or Accuracy?** With Randall Jackson and Caroline Welter.

Research Software

Seedgraph, 2026-present. Python prototype for LLM-assisted research using scholarly citation and concept graphs with source-linked evidence. Command-line, web, and MCP interfaces. Used in personal research and shared with other researchers for feedback. Repository.

Technical Skills

R, Python, MATLAB, Julia, ArcGIS, Git. Libraries include tidyverse, ggplot2, Plotly, caret, XGBoost, data.table, Arrow, NumPy, pandas, scikit-learn, and Matplotlib.

Additional Professional Experience

Testing and Reporting Specialist, Fidelity Investments, Covington, Kentucky, March 2005-January 2014.

Teaching

George Washington University, Department of Economics

- Time Series Econometrics, graduate course, 2020-present.
- Probability and Statistics for Economists, graduate course, beginning spring 2026.

Both courses serve the applied master's program in economics. Across 117 student evaluations of Time Series Econometrics, 109 awarded the highest overall rating (5/5).

University of Cincinnati, Department of Economics

- Intermediate Microeconomic Theory, undergraduate course, fall 2016.
- Introduction to Microeconomic Theory, undergraduate course, fall 2015.

Invited Seminars

- **August 2026:** U.S. Bureau of Labor Statistics. "The Seasons They Are A-Changin': A Century of Definitions and a Way Forward."
- **February 2026:** American University. "It's About Time (Series): A Simple Correction for Difference-in-Differences."
- **February 2026:** University of Cincinnati. "It's About Time (Series): A Simple Correction for Difference-in-Differences."
- **2024:** George Washington University. "Nowcasting Distributional National Accounts for the United States: A Machine Learning Approach."
- **2024:** Murray State University. "Testing a Hypothesis with Multiple Statistics: A Copula-Based Approach."
- **2023:** Federal Housing Finance Agency Seminar Series. "For What It's Worth: Measuring Land Value in the Era of Big Data."
- **2023:** United States Department of Agriculture Seminar Series. "For What It's Worth: Measuring Land Value in the Era of Big Data."
- **2022:** University of Cambridge. "For What It's Worth: Measuring Land Value in the Era of Big Data."
- **2021:** University of Auckland. "Roots from Trees: A Machine Learning Approach to Unit Root Detection."
- **2020:** University at Albany. "The Dark Side of the Moon: Searching for the Other Half of Seasonality."
- **2020:** George Washington University. "The Dark Side of the Moon: Searching for the Other Half of Seasonality."

Selected Conference Presentations, 2025-2026

- **2026, scheduled:** NeurIPS Creative AI Track. "Whose Hand Do You Prefer? The Steerability of Machine Aesthetic Judgment." Accepted; presenter. Date and time to be assigned.
- **May 2026:** North American Econometric Society Summer Meetings. "The Seasons They Are A-Changin': A Century of Definitions and a Way Forward."
- **May 2026:** North American Econometric Society Summer Meetings. "It's About Time (Series): A Simple Correction for Difference-in-Differences."
- **2025:** American Economic Association. "Nowcasting Distributional National Accounts for the United States: A Machine Learning Approach." Selected for *Papers and Proceedings*.
- **2025:** American Economic Association. "When the Levy Breaks: How Failing Renewal School Levies Impacts Housing Prices."